INCH-POUND

MS27997M
26 OCTOBER 1993
SUPERSEDING
MS27997L
31 March 1992

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 25 AMPERES, 3 PST, N.O. TYPE I, HERMETICALLY SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-6106.

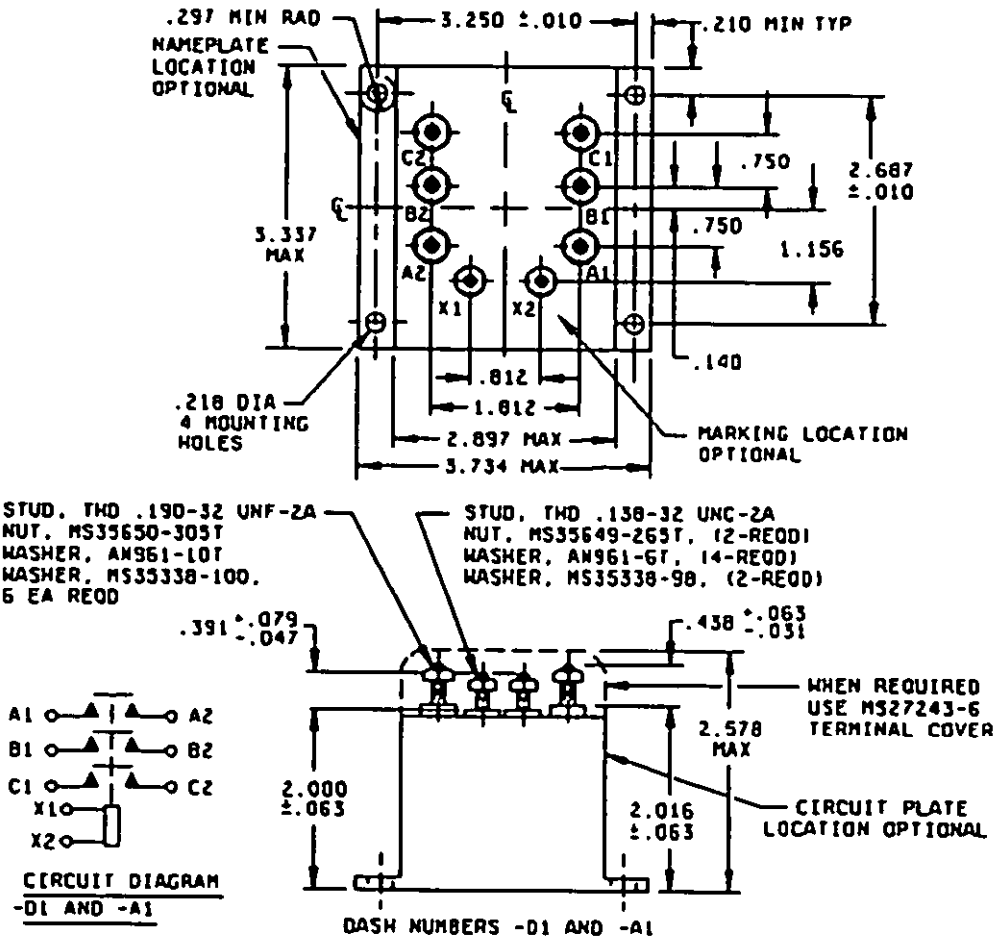


FIGURE 1. Dimensions and configuration.

(M) denotes changes

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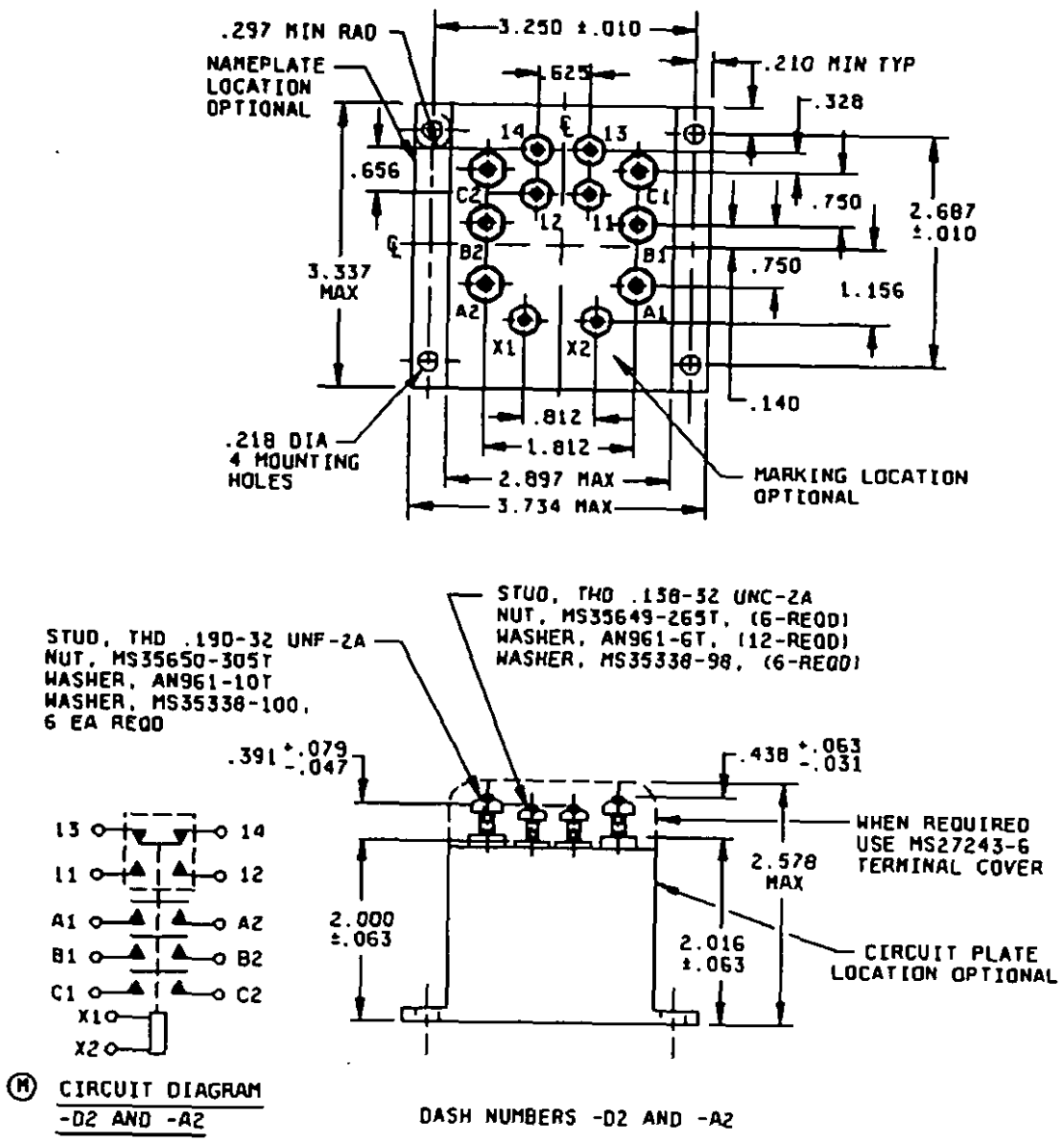


FIGURE 1. Dimensions and configuration - Continued.

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Inches	mm	Inches	mm	Inches	mm
.010	0.25	.297	7.54	1.812	46.02
.031	0.79	.328	8.33	2.000	50.80
.047	1.19	.375	9.52	20167	51.21
.063	1.60	.391	9.93	2.578	65.48
.079	2.01	.438	11.13	2.687	68.25
.138	3.51	.625	15.88	2.897	73.58
.140	3.56	.656	16.66	3.250	82.55
.190	4.83	.750	19.05	3.337	84.76
.210	5.33	.812	20.62	3.734	94.84
.218	5.54	1.156	29.36		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
4. Coil and auxiliary terminals may use additional washer for terminal seat.
5. Terminal covers and barriers may be required at power or auxiliary terminals.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Dash numbers and general characteristics: See table I.

Contact data:

Load ratings: See table II.

Maximum contact drop, initial: 0.150 V.

After life test: 0.175 V.

Overload current (NO): 200 amperes.

Rupture current (NO): 250 amperes.

Coil data: See table III.

Duty rating: Continuous.

RFI specification: MIL-STD-461 (applicable to coil circuits of ac operated relays).

Electrical data:

Minimum insulation resistance:

Initial: 100 megohms.

After life or environmental test: 50 megohms.

Dielectric strength:

Sea level, 2-5 seconds:

	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case:	1,250	1,500	1,000	1,125
Aux. contacts:	1,250	1,500	1,000	1,125
All other points:	1,250	1,800	1,000	1,350

Altitude, 1 minute:

	28 V dc	115 V ac
Coil to case:	500	500
Aux. contacts:	500	500
All other points:	700	700

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Environmental characteristics:

Temperature range: -70°C to +125°C.

Maximum altitude rating: 80,000 feet.

Shock, g-level: 50 g's.

Duration: 6-9 ms.

Maximum duration contact opening: 2 ms.

Vibration, sinusoidal: See table IV.

Nonoperate:

G-level: 15 g's.

Frequency range curve: 20 to 2,000 Hz.

Acceleration: 15 g's.

Terminal strength (high temperature pull and torque test): Not applicable.

Part or Identifying Number (PIN): MS27997- (plus applicable dash number from table I).

(M) TABLE I. Dash numbers and general characteristics.

MS PIN-MS27997-	Type	Coil type	Terminal type	Mounting or mating socket	Auxiliary terminals	1/ Max weight in pounds
D1	I	dc	Stud	Flange	None	1.20
D2	I	dc	Stud	Flange	Yes	1.25
A1	I	ac	Stud	Flange	None	1.25
A2	I	ac	Stud	Flange	Yes	1.30

1/ Weights include covers and barriers.

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TABLE II. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	50	25		5	5	25		5		25				
Inductive	10	25		5	5	25		5		25				
Motor	50	25				25				25				
Lamp	50			1	1			1						
Transfer load														2/
Mechanical life reduced current	100	6.3		1.25	1.25	6.3		1.25		6.3				
Intermediate current	50	Applicable according to MIL-R-6106												

1/ Absence of value indicates relay is not rated for three phase applications.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.

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TABLE III. Operating characteristics.

PIN MS27997-	Coil data											Time (milliseconds maximum)					
	Coil	Rated			Max		Max pick-up voltage			Hold voltage 2/	Drop-out voltage 2/	Operate 3/	Release 4/	Contact bounce 3/			
														Main		Aux	
		Volts 1/	Freq. Hz	Res Ω ±10%	Volts	A	Normal 2/	High temp test	Cont current test					NO	NC	NO	NC
D1	X1,X2	28	dc	51	29	.6	18	21	22.5	7.0	1.5	25	10	2			
D2	X1,X2	28	dc	51	29	.6	18	21	22.5	7.0	1.5	25	10	2		4	4
A1	X1,X2	115	60/400	N/A	124	.2	90	100	104	40	10	30	50	2			
A2	X1,X2	115	60/400	N/A	124	.2	90	100	104	40	10	30	50	2		4	4

1/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With rated coil voltage.
4/ From rated coil voltage.

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TABLE IV. Vibration levels (sinusoidal).

PIN MS27997-	Frequency			
	5-10 Hz	10-55 Hz	55-750 Hz	750-1500 Hz
01	.08 DA	.06 DA	15 g's	12 g's
02	.08 DA	.06 DA	10 g's	10 g's
A1	.08 DA	.06 DA	15 g's	12 g's
A2	.08 DA	.06 DA	10 g's	10 g's

CONCLUDING MATERIAL

Custodians:
Navy - AS
Air Force - 85

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-0902-03)

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